JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

DEPARTMENT OF CIVIL ENGINEERING

SURVEYING & GEODETIC SCIENCE DIVISION

★ PRACTICAL FIELD WORK MANUAL & OBSERVATION LOGBOOK ★

Surveying Field Practice & Total Station Lab Manual

Course Code:	R23-CIV-LAB1	**Academic Scheme:**	I B.Tech II Semester (R23 Scheme)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Field Experiment 1: Chain & Compass Surveying

Ranging and measuring lines across obstacles, Radiation and intersection compass traversing of an open plot.

Field Experiment 2: Differential & Fly Levelling

Bench mark transfer, Profile levelling of road alignment, Inverted staff reading for ceiling benchmarks.

Field Experiment 3: Theodolite Angular Measurements

Repetition and reiteration methods for measuring horizontal angles with high precision.

Field Experiment 4: Curve Setting by Rankine's Method

Setting out a circular curve of radius 150m using transit theodolite and steel tape.

Field Experiment 5: Total Station Area & Coordinate Mapping

Setting up prism reflector, Stakeout procedure, Topographic surveying of campus quadrangle, Data export to DXF.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Field Notebook & Records	20 Marks
Viva Voce & Field Test	30 Marks
External Practical Exam	50 Marks (Total: 100 Marks)

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM